GROUP PROJECT PART FOUR

NOMAD

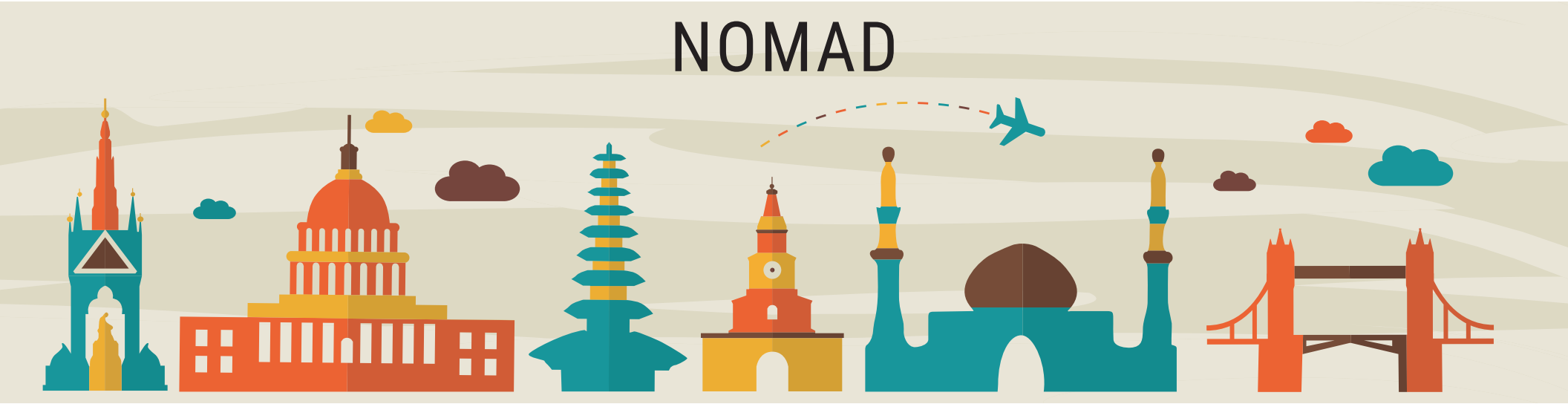

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ILLUSTRATION COURTESY FREEPIK.COM

Nomad takes the stress out of planning group trips. This smartphone app will enable travelers to collaborate on all aspects of a trip, from drafting a trip agenda to documenting the adventures that happen along the way. The app, recommends destinations, supports collaborative itinerary creation, and organizes moments captured during the trip to a centralized destination.

The app's intended users are enthusiastic about using social media and sharing their travel experiences online. They prefer to travel off the beaten path in small groups, as opposed to large tour groups that often only catch glimpses of major tourist destinations. The users are organized and detail-oriented, and they form their plans based on their own interests, the wants of their travel partners and the recommendations of others.

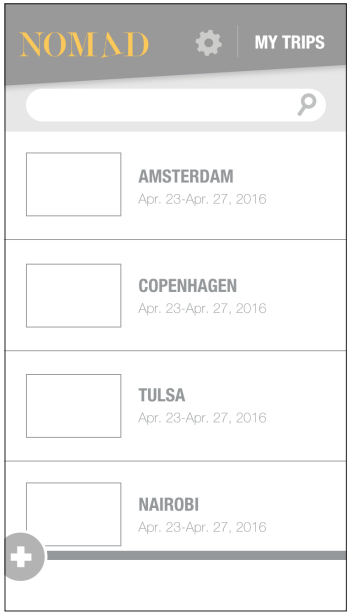
REQUIREMENTS SUMMARY

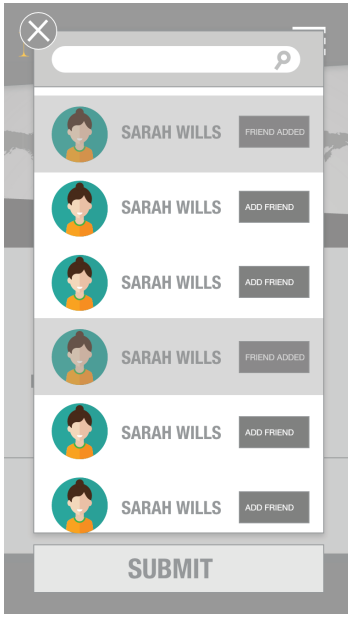
Our application aims to improve group travel experience by focusing on collaboration. In order to fully utilize the features of our application that range from co-creating a shared itinerary to sharing photographs on social media platforms, the application requires access to following:

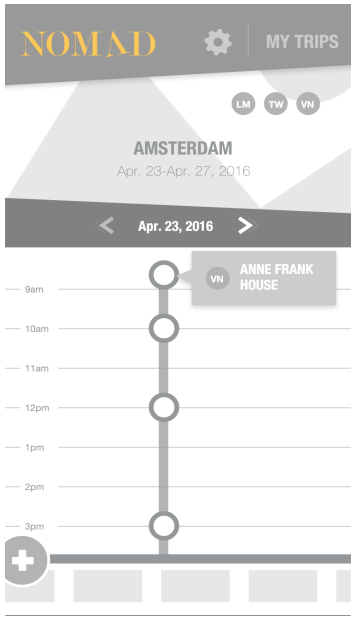
1. Smart phone (Android or iOS)
2. Data Connection (Wi-Fi/Cellular)
3. Login Information

PROTOTYPE REVISIONS

We refined our application significantly, primarily by streamlining the application flow. Specifically, we added a bottom navigation bar that included quick links to the app's core features (trip building, itinerary creator, chat and photo bank). Furthermore, we changed the information archi-







LEFT TO RIGHT
Fig 1.1 Previous landing page; **Fig 1.2** Previous access to photobank; **Fig 1.3** Previous inviting friends screen; **Fig 1.4** previous itinerary screen

tecture, specifically the layout for multiple views such as creating new trip, inviting friends and sharing photographs. We also identified and eliminated trap states in the app by providing users with necessary navigation mechanisms to move around the application and explore all existing features.

In addition to the abovementioned changes, we added new features to our application that reinforce collaboration among group travellers. For example, we added a chat feature that allows people traveling together to discuss ideas and make changes to the itinerary. We also added a recommendations feature. Most of the changes stemmed from our in-class evaluation, where we received constructive feedback from our peers to improve our application holistically.

Landing Page (Fig 1.1.)

The landing page no longer looks like this. We have decided that the new landing page needs to include the user's account/profile at the top of the screen. Also, we decided to change the layout. Our new layout is structured so that a user can search for a destination with a search bar or within one of the categories such as recommended trips. Then once a user clicks on a destination, it will show some fun facts about the destination selected.

Access to Photobank (Fig 1.2)

The photobank is no longer a "scrolling bar" on the bottom of the screen.

Inviting Friends (Fig 1.3)

We have decided to change the layout of this screen for the next prototype. Instead of selecting friends using buttons, we used checkboxes so that the user can select or deselect as many friends as he or she wants before hitting submit.

Itinerary (Fig 1.4)

After our class evaluations, we realized that this itinerary page was quite confusing. In our new design we make it easier for the user to distinguish between the

individual itinerary and the itineraries of friends on the trip.

Design Rationale

As an overview, our changes were aimed towards achieving a streamlined application flow and enhancing the overall user experience. We primarily pushed toward reducing the cognitive load required to complete tasks by contextualizing information wherever applicable. Below are couple of examples:

Creating a New Trip

Instead of presenting users with information they need to enter all at once in a single screen, we decentralized this task by creating multiple screens, which allows the user to focus one section at a time and thereby reducing their cognitive load.

Inviting Friends

By modifying the layout and interaction pattern from button click to checkmark style, we increased the feedback provided to a user by displaying names of friends selected thereby allowing larger group selections. Furthermore, by incorporating the new interaction pattern, we reduced the amount of cognitive memory a user needs to recollect and confirm invited friends. This minor change is significant because we realized inviting friends to a trip is a critical touchpoint in our application, as an error can result in creating friction among people which degrades user experience.

Adding Items to the Itinerary

We retained the core look of the itinerary section but made some minor adjustments so primary elements — such as the "add item" button — would better stand out to users. We moved the "add item" button from the left to the right and pushed it away from the bottom bar. Additionally, instead of looking like a generic plus sign, the button is now features plus sign inside of a pinpoint shaped icon. We also moved the users' profile pictures closer to the itinerary and matched the border around their photo to their individual itinerary paths so users could better understand who was going where and when. The itinerary lets users toggle on and off other users' paths so the screen can be as busy or as clean as an individual prefers. Finally, we simplified the itinerary creation process by letting users add their own item or select activities or places from categories. Before, users had to type in a name, time, location and description, as well as upload an image. Now, users can quickly select an activity, scroll to the time they prefer, and the app will take care of the details like location. The app also automatically adds images to items selected from the categories.

Photo Bank

By eliminating the "scrolling" horizontal photo bank feature on the bottom of the screens, we have made the task of uploading/sharing a photo less confusing. We replaced the horizontal photobank with a navigation bar that has a "photobank" item that users will, for the most part, be able to access throughout the app. Including a camera icon that links to the photobank screen on the navigation bar is much more intuitive and follows standard conventions.

EVALUATION

Evaluation techniques and procedures:

Before testing the high-fidelity prototype with test users, team members completed a cognitive walkthrough of the app. This method is important because it allowed us to identify and inspect any existing issues with the interactive parts of the app. We wanted to determine how easily users could accomplish the tasks that we assigned them.

We tested the high-fidelity prototype with five users, all of whom considered themselves to be frequent travelers. Unlike an earlier survey, in which we asked users to quantify the frequency of their trips, we did not ask these test users specifically how often they traveled. Instead, we let them determine themselves whether or not they thought they were a frequent traveler and one who might use travel-planning apps or websites.

Team members split up to complete the usability tests. For each test, the team member read through a script that explained the purpose and goals of the test, then asked if the participant had any questions (See Appendix A). After that, the participant began taking the usability test and attempted to complete the following four tasks:

1. **Scenario One:**
   Since your graduation 4 years ago, you have been working in San Francisco. It's been long since you've met your friends and all of you are planning for a reunion in Paris. Use the nomad application to:
   a. Task One: Create a new trip to Paris from May 1 to 6, 2016.
   b. Task Two: Invite your friends Debbie, Katie and Matt to the trip.

2. **Scenario Two:**
All of you are enthusiasts when it comes to visiting museums and can spend the whole day discussing about the same. There's unveiling of a new artwork by Canson at the Louvre Museum and all of you are excited. Use the nomad application to:

   a. Task Three: Add Louvre Museum to the itinerary on May 1, from 2 p.m. to 4 p.m.

2. **Scenario Three:**
After six exciting days in Paris, you are home in San Francisco and cannot wait to share those amazing moments and beautiful sceneries with their family and friends. Use the Nomad application to:
   a. Task Four: Share the first four photographs on Facebook.

Team members asked participants to verbalize their thoughts and actions as they were completing the tasks. The think-aloud method is valuable because it helps the team get a sense of why users tended to click certain page elements or where they ran into confusion.

Following the test, task administrators sat down with participants for an informal, open-ended conversation about the prototype. The conversation served as a way to jog the participant's memory about interacting with different aspects of the app. Participants also took an online survey (Appendix B)that prompted them to reflect on the experience. We used the survey because we recognize that some people are not comfortable with sharing their thoughts directly with the test administrator. This way, they can anonymously provide feedback to the team.

Evaluation Results:

Overall + Home screen

In general, users liked the look and feel of the design. One of the participants did not like the color scheme, saying there was too much gold; however, other users said they found that the colors were bold and inviting. One user said she thought it looked easy to use and that it made her want to plan a real trip. The same participant also liked the photography featured throughout the app.

One user felt that the home screen made her feel as though she had a limited number of travel destinations from which she could select. She saw the search bar but wondered if smaller towns and villages would show up — and if they didn't, would the app be useless? (It wouldn't — you just would not receive any recommendations). She said she wants to see variety, and she wouldn't use the app unless she knew she could use it to plan any trip she could imagine.

Task 1 - Create trip

Users were split on the trip recommendations feature. Some liked the idea of being able to explore their options while others said they and their friends usually go into planning a trip with a location already determined. Still, the recommendations did not seem to discourage anyone from using the app.

When one user clicked on the link to Paris, he noticed the picture and text and thought, "This is a general introduction to Paris, but I don't need an introduction."

Task 2 - Add friends

We included a list of favorites/recent friends at the top of this section, assuming that users would see these people first and click them. We did not anticipate that they would look straight to the list of names underneath. Users were slow to select names from the list, often clicking on the photos or names first instead of the checkbox. "Why can't I select these people just by clicking on their names?" one user asked.

Task 3 - Add item to itinerary

All users found the "add item" button without much searching, particularly compared to the users who tried our low-fidelity prototype. One user found that the multi-path itinerary was too complex, but other users said they liked being able to see where their friends were going to be in case they wanted to plan a few activities on their own and meet up later.

Task 4 - Upload photos from photo bank

"Sharing photo is almost mandatory now," one user said. This user found that this feature would be useful for his group of traveling friends because they like to share pictures among each other anyway, as well as online.

Most users liked the concept of the photobank and did not have trouble uploading the images. One participant commented that the "select" button in the top right corner, which is yellow, is hard to see and that she almost missed it the first time she looked for it. But, she added, select is in the same place as on the general iPhone photo app, and that increased the memorability of this app's interface for her.

One user said the photobank tool wasn't necessary. "I think it will cause more issues than good," she said. The user said she primarily uses her iPhone camera app and uploads photos to Facebook directly from the phone's camera roll. She thought the photo map was a nice touch but didn't picture herself using it in practice.

Recommendations

Users want Nomad to not only help them plan a trip but also help them figure out what trip to take. We had already introduced the idea of recommended trips, but one user suggested we take it a step farther by customizing the recommendations for each user. "On Netflix, if you like a movie, you'll get recommendations for other movies you might enjoy," she said. "I want an app that expresses the types of trips I like. I don't like cruises, so I wouldn't want to open up the app and see recommended cruises."

The same user also recommended other potential user profile features, such as adding one's hometown so the app could base trips around nearby transportation hubs. "It would be cool if I had, say, five days in July, the app would tell me what are some good places to go, maybe from a weather standpoint and a financial standpoint," she said. The user said this kind of customization and personalization would make her more likely to switch to Nomad from existing travel apps like TripAdvisor.

Though we plan for Nomad to have some offline capabilities, we recognize that some features, like chat, will require Internet access. One user suggested forcing certain features to not work unless you're connected to wifi. "If one person is updating the itinerary from the hotel, you're not wasting data because it won't notify you of these updates," the user said.

Final proposed changes:

Issue 1: Personalized recommendations

Feedback from multiple users in the usability study group suggests we need a personalized new trip suggestion system in the landing page. This feature would give users personalized trip recommendations that are based on their past trips, places, searches and personal preferences.

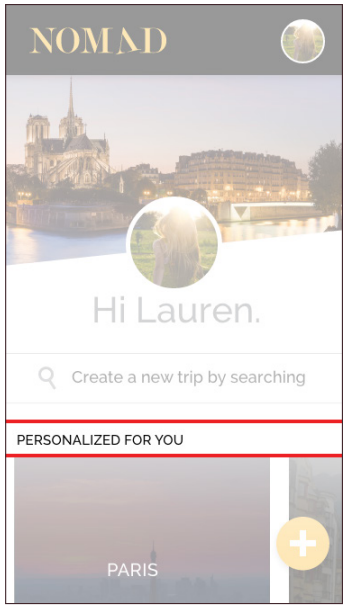
Issue 2: Provide weather and price informations

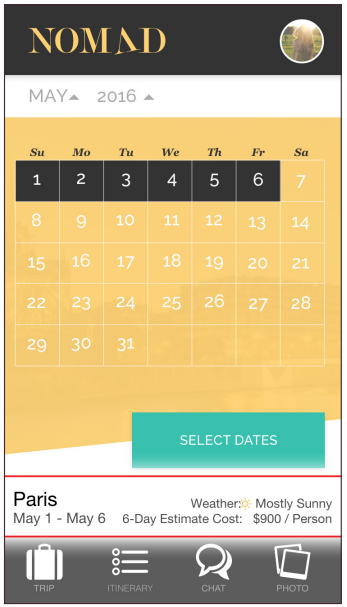
During the usability testing sessions, one user mentioned including typical weather and price information. We think this feature is essential for travelers to make decisions, and weather information and estimated cost are provided in the date selecting page to helps the user decide if the date is suitable for their trips and personal preferences.

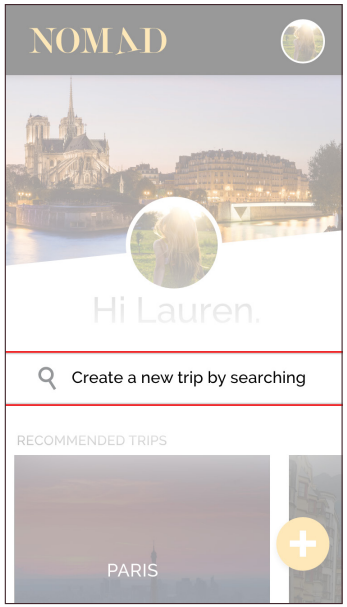
Issue 3: Indication for both recommendation and search in home screen

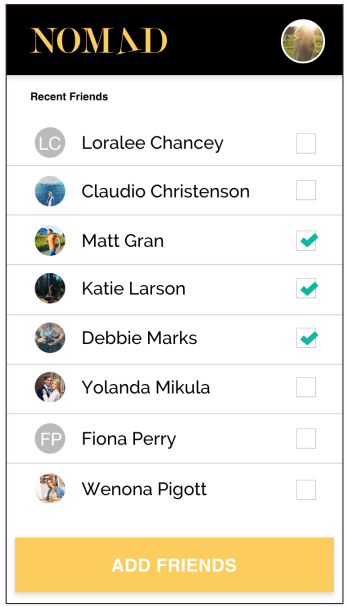
Two users mentioned they didn't know if they had to choose a trip from one of the below selected trips. Here, the text in the search bar is changed from "Search for a new adventure" to "Create a new trip by searching." This change indicates users they can create a trip by either searching or browsing for recommendations.

LEFT TO RIGHT
Fig 1.5 personalization; **Fig 1.6** Weather and cost estimate; **Fig 1.7** Searching for a new trip;
Fig 1.8 Friend screen









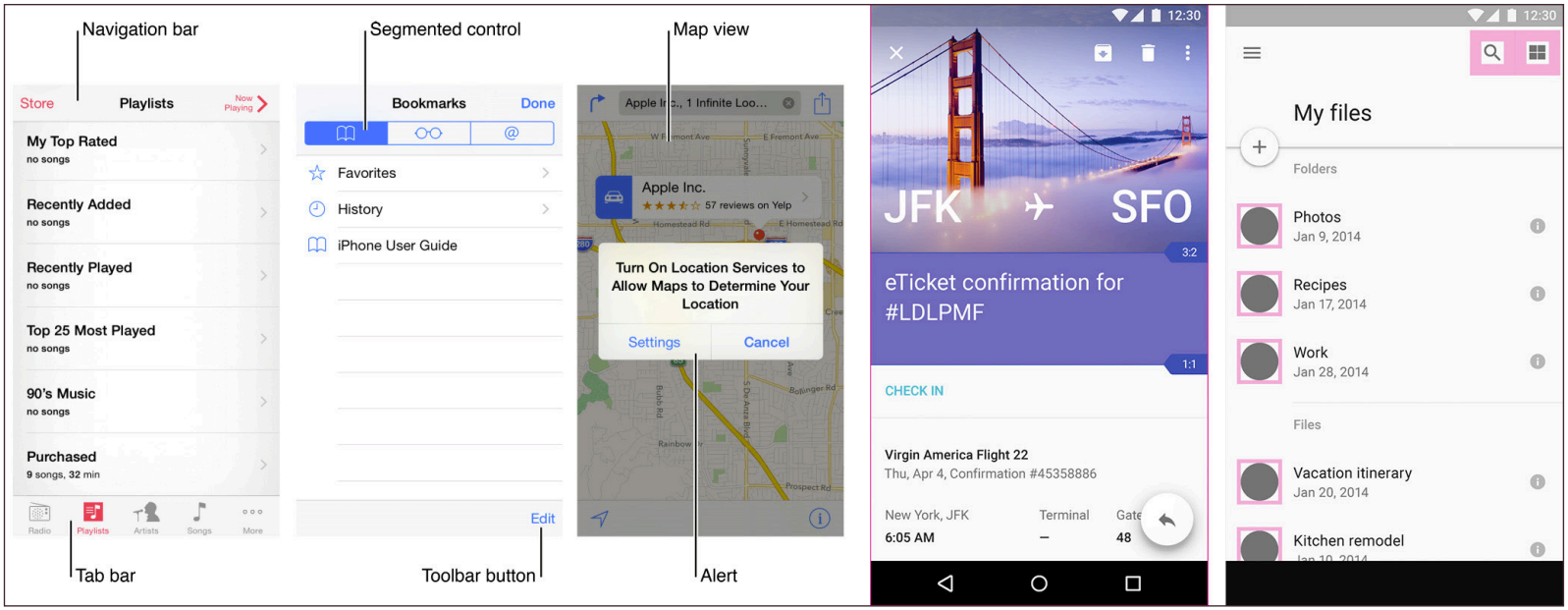


Fig 1.9 Example Illustrations from iOS and Android Design Guidelines

Issue 4: Friends list has inconsistent profile pics and names

All users in our usability testing group were confused by this add friends page. The future design, as shown above, removes the recents friends section on top and allows users to instead scroll down a list and click the names of friends or checkboxes next to them to select.

Issue 5: Color and Typography

Example Illustrations from iOS and Android Design Guidelines

Color was identified as an important issue by our final evaluation user group. The feedback suggests Nomad mobile app should adopt design guidelines for different platforms: Google Material Design Guidelines and iOS Human Interface Guidelines.

Design

Based on our initial feedback, in the future, we would use a more collaborative tool such as Github so that everyone could work simultaneously on different parts of the design and merge our changes into one final product. Likewise, we could have spent more time researching better prototyping software tools to use. For this project, we used Adobe XD to design all of our prototypes. Adobe XD was fun to try, but its functionality was still very limited since the software is still in the beta stage. It didn't allow us to do everything we hoped to be able to do. We had to cheat the system to get around some constraints (stretch the screen to simulate vertical scrolling) or give up an idea entirely (horizontal scrolling to view recommended trips). One mistake we didn't realize until after beginning the process of building our prototype is that it may have been more beneficial and efficient for us to sketch out our the interface connections earlier. We sketched out general features and functionality that we agreed should be included in the application. However, we didn't sketch out the complete flow of the app until after we designed a handful of screens, and then we realized we had overlooked a few holes in the design.

Evaluation Techniques

For future work, we plan on recording our user sessions using usability testing software such as Lookback, Morae or Userzoom. These tools will give us rich data such as the exact time it takes a user to complete the task and facial expressions and audio feedback from the user as he or she is completing a task. Although a human observer is able to do these things, we have to account for human error. There may be things that the observer or notetaker doesn't notice that a powerful usability testing tool would be able to pick up on. Or sometimes, users may not express something verbally, but his or her facial expression says everything. These are just some examples when a usability testing software will be beneficial for a usability test.

The pool of people who tested the final prototype (five) was much smaller than the number of respondents we received from our surveys (67). We plan on testing a larger group of users in the future to get even more feedback on how we can improve our design and the usability of our system. Our design can certainly be more polished and flow much more smoothly. However, we will need a diverse and larger group of users to test out our product to point out flaws in our design.

Furthermore, it would have been beneficial for our team to do a cognitive walkthrough earlier in the design process to evaluate some of our initial designs more quickly. By doing this, we would have picked up on some of the issues in our design that may inhibit or prevent a user from completing a task.

APPENDIX

APPENDIX A — EVALUATION SCRIPT + TASKS

Test Script

Hi, <participant name>. My name is <test administrator name>, and I'm going to be walking you through this session today.

Before we begin, I have some information for you, and I'm going to read it to make sure that I cover everything. You probably already have a good idea of why we asked you here, but let me go over it again briefly.

Nomad takes the stress out of planning group trips. This smartphone app will enable travelers to collaborate on all aspects of a trip, from drafting a trip agenda to documenting the adventures that happen along the way. Today, we're asking people to try using our application that we're working on so we can see whether it works as intended. The session should take about 30 minutes.

The first thing I want to make clear right away is that we're testing the application, not you. You can't do anything wrong here. In fact, this is probably the one place today where you don't have to worry about making mistakes. As you use the app, I'm going to ask you as much as possible to try to think out loud: to say what you're looking at, what you're trying to do, and what you're thinking. This will be a big help to us. Also, please don't worry that you're going to hurt our feelings. We're doing this to improve the application, so we need to hear your honest reactions.

If you have any questions as we go along, just ask them.I may not be able to answer them right away, since we're interested in how people do when they don't have someone sitting next to them to help. But if you still have any questions when we're done I'll try to answer them then. And if you need to take a break at any point, just let me know.
Do you have any questions so far? (If yes, note them now and answer accordingly.)
OK, great. We're done with the questions, and we can start looking at things.
First, I'm going to ask you to look at this application and tell me what you make of it:

1. What strikes you about it,
2. What you can do here, and what it's for.
3. Just look around and do a little narrative. You can scroll if you want to, but don't click on anything yet.

Allow this to continue for one or two minutes, at most.

Thanks. Now I'm going to ask you to try doing some specific tasks. I'm going to read each one out loud. And again, as much as possible, it will help us if you can try to think out loud as you go along.

Hand the participant the first scenario, and read it aloud. Allow the user to proceed until you don't feel like it's producing any value or the user becomes very frustrated.

Tasks

Scenario One

Since your graduation 4 years ago, you have been working in San Francisco. It's been long since you've met your friends and all of you are planning for a reunion in Paris. Use the nomad application to:

- Task One - Create a new trip to Paris from May 1st to 6th, 2016.
- Task Two - Invite your friends Debbie, Katie and Matt to the trip.

Scenario Two

All of you are enthusiasts when it comes to visiting museums and can spend the whole day discussing about the same. There's unveiling of a new artwork by Canson at the Louvre Museum and all of you are excited. Use the nomad application to:

- Task Three - Add Louvre Museum to the itinerary on May 1st, from 2pm - 4pm

Scenario Three

After six exciting days in Paris, you are home in San Francisco and cannot wait to share those amazing moments and beautiful sceneries with their family and friends. Use the Nomad application to:

- Task Four - Share the first four photographs on Facebook.

Closing + Survey

That completes our tasks for today. If you could help by completing a short questionnaire. bit.ly/PostTestNomad

Do you have any questions for me, now that we're done? Thanks <participant name>, for taking time and providing valuable feedback. I appreciate the help!

APPENDIX B — POST-TASK QUESTIONNAIRE

All information collected will be used productively in improving usability of the system. Please support us by being completely honest when answering the following questions.

1. Describe your first impressions when you viewed the system.
2. On a scale from 1-5 with 1 being the least and 5 being maximum, how confident were you about completing all the tasks, after spending 5 minutes using the system?
3. While you were completing the tasks, list specific instances when you weren't sure about your actions.
4. Given a chance, how would you improve the system?
5. Describe in a few sentences your overall experience while using the system.

OUR PROTOTYPE CAN BE VIEWED AT: HTTP://ADOBE.LY/26ZPEMF